

PROCESS MODELS – NAUCOM

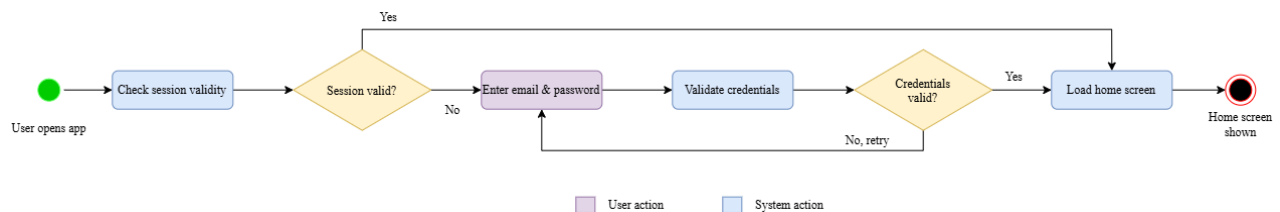
NauCom’s process logic is documented at two levels, each for a different audience. This doc shows both, side by side, and explains how they connect.

Level	Model	Audience	Shows
Business	BPMN	Stakeholders, non-technical reviewers	What decision gets made, and why
Technical	Activity Diagram	Developers	How the app/backend actually implements that decision

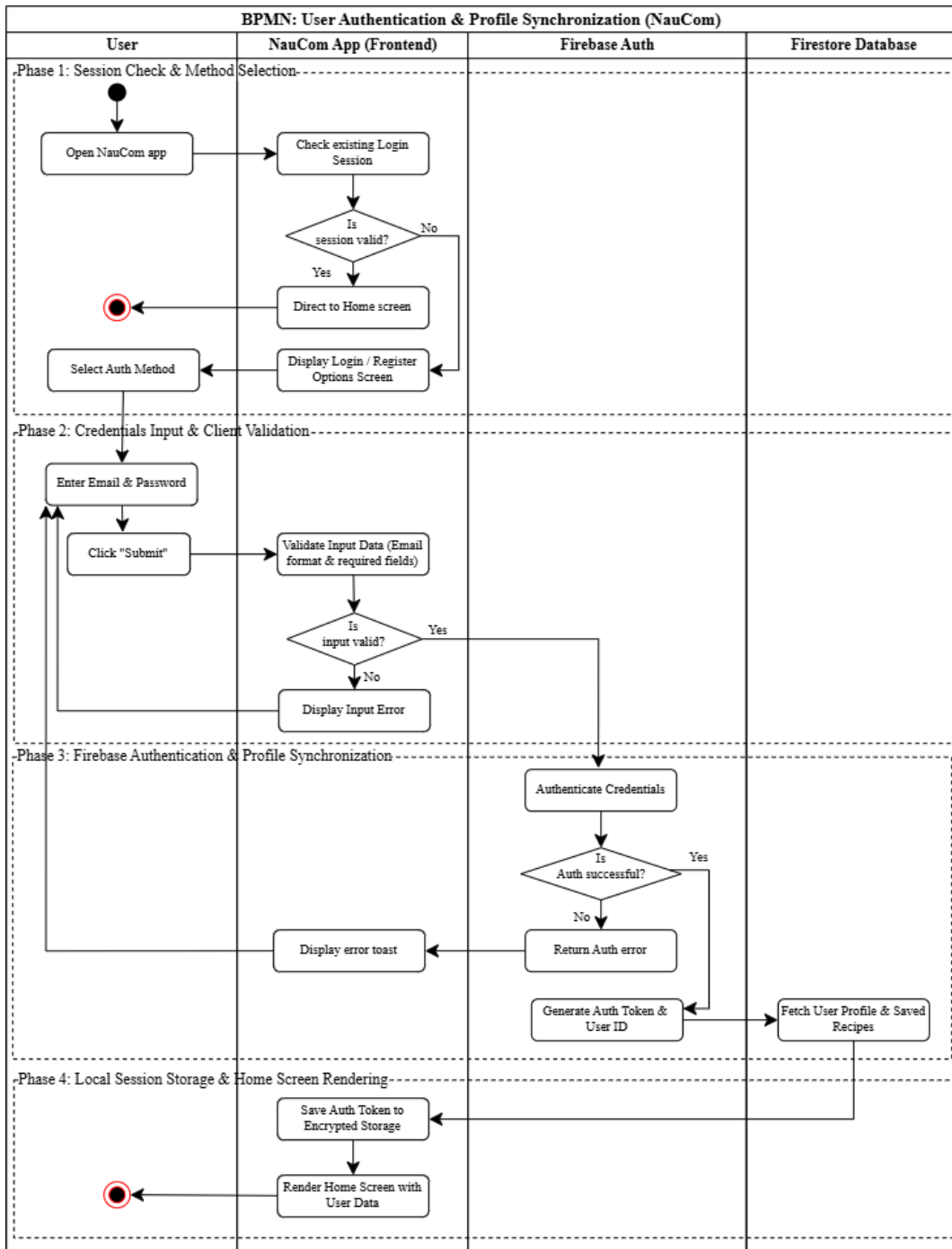
Every BPMN task below expands into a more detailed sequence in its matching Activity Diagram - the mapping tables under each pair show exactly which technical steps a given business-level task stands in for.

1. User Authentication

Business Process (BPMN)



Technical Flow (Activity Diagram)



How they connect

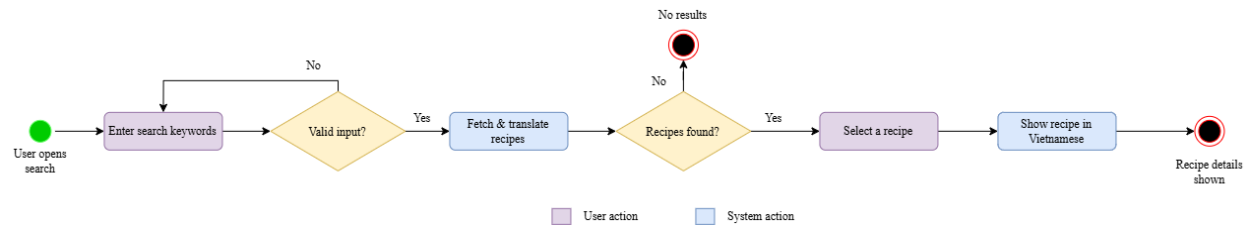
BPMN task (business level)	Expands into (Activity Diagram)
Check session validity	Phase 1: Check existing login session, direct to Home screen if valid
Enter email & password	Phase 1-2: Select Auth Method → Enter Email & Password → Client-side input validation

Validate credentials	Phase 3: Authenticate Credentials via Firebase Auth
Credentials valid? (No → retry)	Phase 3: Auth successful? No → Return Auth error → Display error toast
Load home screen	Phase 3-4: Fetch User Profile & Saved Recipes → Save Auth Token to Encrypted Storage → Render Home Screen

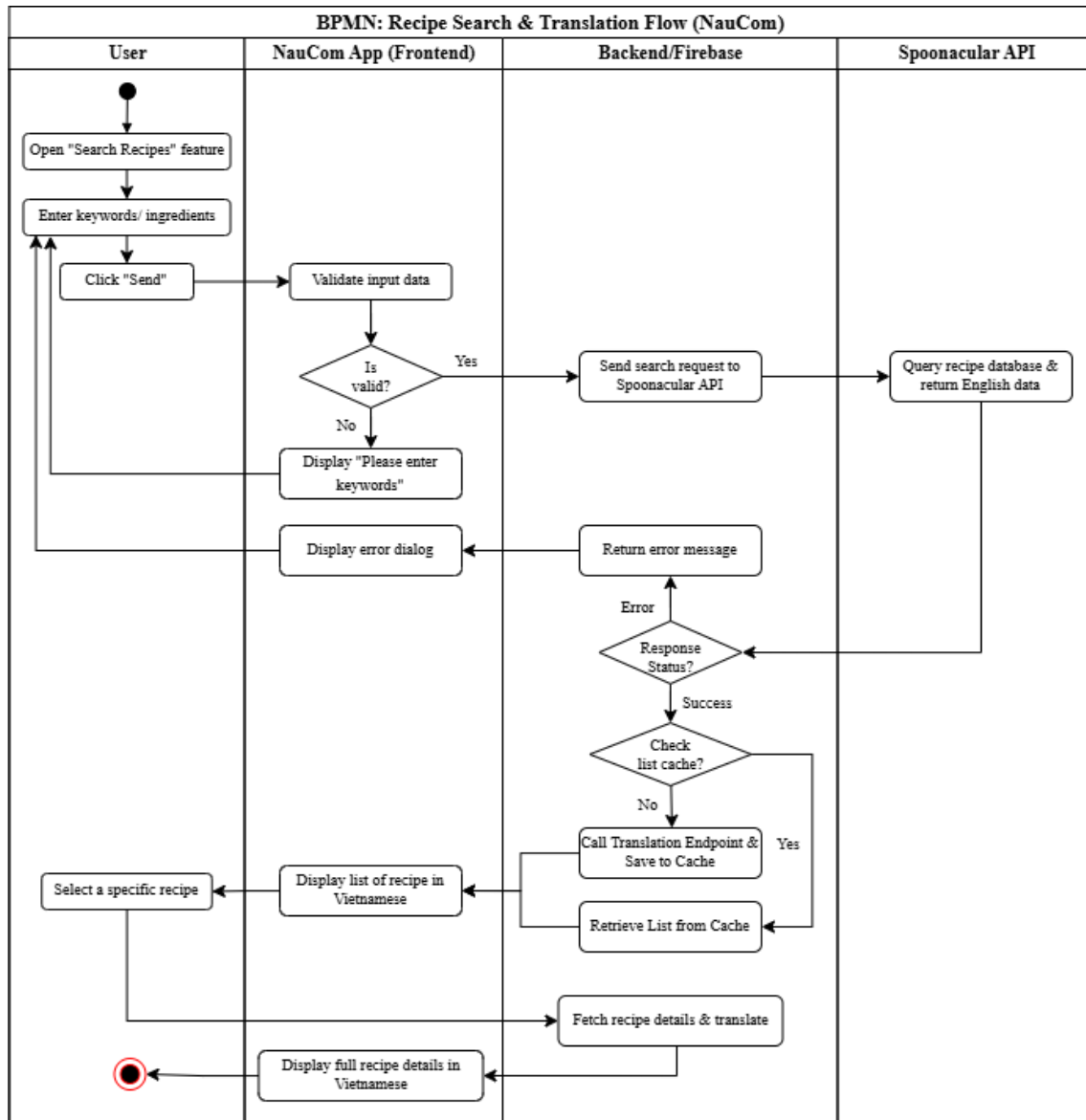
The BPMN treats "Load home screen" as one business outcome; the Activity Diagram shows it actually requires three separate technical steps (token storage, profile fetch, rendering) — detail a stakeholder doesn't need, but a developer does.

2. Recipe Search & Translation

Business Process (BPMN)



Technical Flow (Activity Diagram)



How they connect

BPMN task (business level)	Expands into (Activity Diagram)
Enter search keywords	Enter keywords/ingredients → Click "Send"
Valid input? (No → retry)	Validate input data → Is valid? → Display "Please enter keywords"
Fetch & translate recipes	Send search request to Spoonacular API → Query recipe database → Check list cache → Call Translation Endpoint & Save to Cache (or Retrieve List from Cache)
Recipes found? (No → end)	Response Status? → Error → Return error message → Display error dialog

Select a recipe	Select a specific recipe → Display list of recipe in Vietnamese
Show recipe in Vietnamese	Fetch recipe details & translate → Display full recipe details in Vietnamese

Here the gap is largest at "Fetch & translate recipes" - one business task, but the technical flow underneath includes an API call, a cache check, and a conditional translation step. That caching logic is an implementation optimization with no business-level meaning, so it correctly stays out of the BPMN.

Why two models instead of one?

An earlier version of this documentation used BPMN notation to describe the technical flow directly (system components as swimlanes, cache/token logic as tasks). That conflated the two levels: a diagram meant for stakeholders ended up requiring dev-level context to read, and a diagram meant for developers was mislabeled as a business process. Splitting them means each diagram now answers exactly one question - "what does the business decide?" or "how does the system do it?" - instead of doing both jobs badly.